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XIAOFENG CAO, Wuhan Donghu University, Wuhan, Hubei, China

JINGJING DUAN, VSB – Technical University of Ostrava, Ostrava, Moravian-Silesian Region, Czech Republic

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Xiaofeng Cao
Wuhan Donghu University
Wuhan, Hubei, China
authka@163.com

Jingjing Duan*
VSB - Technical University of Ostrava
Ostrava, Ostrava, Czech Republic
Wuhan Business University
Wuhan, Hubei, China
jingjing.duan.st@vsb.cz

Abstract

The emergence of AIGC offers groundbreaking potential to transform the way digital media, particularly games, create and personalise content. However, there are several key issues that limit the full realisation of AIGC capabilities. Currently, the application of AIGC technology still faces a number of challenges at the company level, such as high technological thresholds, long development cycles, and high implementation costs. In addition, it is difficult for companies to balance resource inputs and effective outputs as AI technology is still immature and difficult to seamlessly integrate it into existing frameworks. Therefore, there is a need to assess the operational efficiency and innovation capabilities of companies attempting to adopt AIGC to identify areas for improvement. In this paper, 20 listed Chinese digital media companies with AIGC as a keyword were screened and their technology innovation efficiency in 2023 was assessed using a data envelopment analysis (DEA) model. It is found that the industry faces multiple inefficiencies: 75% of the sample companies exhibit technological inefficiencies, which implies a mismatch between resource inputs and outputs, usually due to the lack of short-term returns on AIGC technologies that require significant time and investment to develop. In addition, 65% of the companies face technical inefficiencies on the pure technology side, reflecting management and application challenges, such as poor teamwork or inability to successfully apply AI technologies to business scenarios. In addition, 45% of companies suffer from scale inefficiencies, as some are too small to realise synergies, while others are too large, leading to wasted resources and operational inefficiencies. Therefore, this paper highlights the need for comprehensive improvements in technology innovation, management optimisation and scale control to address these challenges and enhance the application of AIGC in digital media to help digital media companies grow healthily.

*Corresponding author.



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CCS Concepts

• **General and reference** → Cross-computing tools and techniques; Evaluation.

Keywords

AIGC, Digital Media Companies, Technological Innovation Efficiency, DEA

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1 Introduction

In the wave of digital media, artificial intelligence is reshaping the boundaries of content creation in its own unique way. Artificial Intelligence Generated Content (AIGC), an emerging technology, has expanded from its initial focus on text generation to the more complex realm of audiovisual language creation [1]. Open AI's Sora grand model, the latest breakthrough in AIGC, is capable of generating high-quality video based on textual cues, marking a significant advancement in AI's approach to audiovisual language creation [2]. The field of AI image processing Stable Diffusion and Midjourney are metamorphosing at breakneck speed and showing their first signs of success [3]. The SD ControlNet model, when introduced, has enhanced its functionality and performance by improving the training dataset and training steps, greatly improving the image controllability rate and stability. The model provides significant improvements to the colour shift problem and can replace canny+openpose under certain conditions.

The application of AIGC greatly expands the creative boundaries of artists and promotes interdisciplinary integration [4]. By assisting in generating art and optimising design, AIGC provides more tools and inspiration [5]. This not only changes the traditional way of creating art, but also opens up endless possibilities for future artists. In digital media, AIGC-driven tools allow creative workers to focus on more creative and strategic tasks, while leaving technical operations to AI, redefining the production model of the digital art industry. In the field of game design, AIGC algorithms are used in character design, scene generation, game difficulty adjustment, etc., making the game much more interactive, intelligent

and playable, and AIGC-driven characters (NPCs) are able to adjust their reactions and strategies in real time according to the player's behaviours, which enhances the immersiveness and challenge of the game.

Games based on AIGC technology can not only convey cultural information through fine art design, but also dynamically adjust the presentation of cultural elements by analysing players' behaviours and preferences, thus enhancing the attractiveness and infectiousness of culture. This feature is particularly suitable for culturally rich countries like China to promote their local culture globally by enhancing cultural identity through games and digital art. In this paper, we select digital media companies with AIGC as the theme word among listed companies in China, construct an input-output indicator system, and calculate the efficiency of scientific and technological innovation with the DEA model. The present situation of the efficiency level of this industry is understood, and countermeasures are proposed to improve the efficiency of science and technology innovation of digital media companies in the context of AIGC application.

2 Literature Review

The application of artificial intelligence in digital media dates back to the 1960s, when artists and computer scientists collaborated to generate simplified images and music using computers [6]. As technology advanced, AI gradually evolved into a powerful creative tool, primarily categorized by function into discriminative AI and generative AI. AI-generated content includes text generation, image generation, audio generation, and video generation [7].

Recent advances in technologies such as Generative Adversarial Networks (GANs) and Deep Learning have significantly broadened the scope of AI applications in the field of digital media. In 2018 the GAN-generated artwork 'Portrait of Edmund Bellamy' was sold for a high price at Christie's auction house, marking the first successful breakthrough of AI art in the mainstream art market.

In the study of AIGC technology, machine learning and computer vision are the core technologies that drive its application in digital media [8]. Machine learning empowers AI to learn and infer autonomously through data training models, helping it to recognise features of multi-dimensional data such as images and sounds. Computer vision, on the other hand, empowers AI to read, process and understand visual information, enabling it to make significant breakthroughs in areas such as image generation, style migration and 3D modelling. For example, through computer vision technology, AI is able to learn and mimic the creative style of human artists, thus generating images with high artistic value in a short time [9]. In game design, AI uses computer vision technology to generate highly realistic virtual characters and scenes, which greatly enhances players' immersion and interactivity.

At present, AIGC technology has been well applied in some well-known art works and games. For example, in art creation, AI artist 'Obvious' team used GAN technology to create a number of works of art with a unique style [10], and these works are widely noticed by collectors. In the field of gaming, Black Myth: Wukong, a Chinese game that has received a lot of attention, has achieved an unprecedented level of character design, scene generation and plot control using AI technology.

3 Application of AIGC technology in the field of digital media

3.1 Comparison and presentation of AIGC models

The AIGC model iterates from GAN to diffusion models to produce algorithms that are powerful and optimised for results. GAN, the base model of the AIGC model, generates images using generators and discriminators that work against each other. VAE further sidesteps adversarial training by using variational inference of the encoder and decoder to construct the image. Diffusion models rely on long Markov chains to generate high quality images. The main application architecture of the three models is shown in Figure 1.

The most widely used AIGC technology, image generative AI, is introduced as an example. The mainstream image-generative AI models in recent years are shown in the table below.

3.2 Examples of the application of AIGC technology in the gaming field

D5 Real-Time Ray Tracing Renderer. After version 2.6, the new AI Ambience Matching is introduced to set a natural and realistic ambience by using a lightweight convolutional neural network to identify features and a comparative learning algorithm for embedded spatial mapping, which is both accurate and efficient in generating ambient parameters that fit the reference map. Atmosphere matching ranges from, but is not limited to, sky effects, weather (e.g. rain, snow, fog, etc.), and overall colour tones, and the AI intelligently determines whether to use geographic skies to replicate natural variations, or to choose the appropriate HDRI to create a unique light and shadow effect [13]. When the project needs to match the same visual effect in different scenes, AI makes use of neural network colour migration algorithms that adapt to the colour space to greatly improve the accuracy of colour matching, so that the final scene hue can be fused with the reference image with high precision, reducing the rate of trial and error. Even in the face of incomplete matching, the end parameters can be manually adjusted to make it fit.

DLSS (Deep Learning Super Sampling) 3.0 technology. The application in Black Myth: Wukong greatly improves the game's graphic performance and performance optimisation. This technology upsamples low-resolution gameplay footage to high-resolution through deep learning algorithms and significantly improves frame rate and operational efficiency without compromising visual quality. The core of DLSS 3.0 is intelligent rendering of each image frame through AI, which allows the game to render scenes at lower native resolutions, but the final output looks as if it was rendered in high resolution. DLSS One of the key innovations introduced in 3.0 is the frame generation technology Frame Generation, which is capable of generating new frames through AI to dramatically increase frame rates. For games like Black Myth: Wukong, which have complex battle scenes and fast action switches, Frame Generation makes the battle process smoother by predicting and generating transition frames between player actions. In Black Myth: Wukong, DLSS 3.0 also combines dynamic resolution scaling and AI anti-aliasing technology. When certain scenes in the game are overloaded on the graphics card due to complexity, DLSS can dynamically adjust

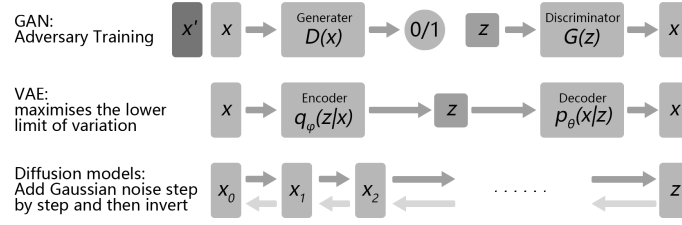


Figure 1: Application architecture of GAN, VAE, Diffusion models [11]

Table 1: Mainstream image generation macromodels

Model Name	Institution	Year	Open source or not	Advantages	Disadvantages
Stable Diffusion [12]	Staility AI	2022	Open source	high-quality and high-resolution images, high controllability and openness, low computational resources	slow generation speed, limited comprehension of complex textual cues, requires quality and diversity of training data
Midjourney	Midjourney	2022	Not open source	unique art style and exquisite details, clean UI , simple and suitable for creative expression	slow generation, not suitable for independent customisation of solutions, less image control
DALL-E	OpenAI	2021	Not open source	generate high quality images from more complex text images, suitable for scene and character fusion	requires high computing resources, limited localised operation, image quality is too high resulting in less controllability
Imagen	Google	2022	Not open source	generate high quality images from more complex text images	requires higher computing resources, more difficult to get started, less controllable images
NVIDIA GauGAN	NVIDIA	2019	Not open source	high-precision conversions from draft to finished drawing, expertise in scenario design	weak image generation for unconventional text, less controllable, more demanding on hardware, less friendly to users with weak draft drawing skills
Pixray	Pixray	2021	Open source	Supports a variety of algorithms, the ability to customise image parameters, support for localised operation, high openness	no clear UI interface interaction ,difficult to get started, insufficient generated images

the resolution to ensure that the game does not drop frames or lag. Meanwhile, AI anti-aliasing technology eliminates jaggies and edge blurring in images through deep learning, making the edges of the screen smoother and improving the overall visual quality.

Wave function Collapse. In the process of generating game maps, the wave function collapse algorithm, which is both regular and random, is used to generate regular types of buildings, tunnels, dungeons, snow, trees, rivers and other maps. The wavefunction collapse algorithm borrows the concept of wavefunction collapse from quantum mechanics, which is the phenomenon of gradually collapsing from multiple possible states to a definite state. In the wave function collapse algorithm a position is determined by a

process called collapse. When a position is collapsed, continuing to randomly collapse neighbouring positions may result in two neighbouring buildings not being able to connect coherently, so in order to ensure the coherence and reasonableness of the entire map need to formulate constraint rules to ensure that the buildings are reasonably connected. And so on, according to these constraint rules whenever a position is collapsed, the available buildings in the remaining positions will be adjusted one by one, so as to ensure that the subsequent collapsed positions can be connected with the collapsed positions, which is the core principle of the wavefunction collapsing algorithm. By repeating the two steps of collapsing and

adjusting the available buildings, all the locations are filled with buildings to get a complete game map.

3.3 Significance of AIGC technology in the field of gaming

AIGC has a wide range of applications and significance in the field of digital media, and its specific significance varies due to the different segments of the industry. Take the game field as an example to analyse the significance of AIGC technology development.

Firstly, it improves the efficiency of game content creation. Generative AI can provide game developers with powerful arithmetic power, which greatly improves the efficiency of game content creation [14]. Through specific procedural means, AIGC can automatically create a number of scene elements such as environment modelling, terrain algorithms, props design, generic vegetation, etc., which conforms to the description, saves the time cost required for repeated manual operations, and increases the completion degree of scene design. At the same time, using the constraint mechanism, AIGC can randomly generate map levels with both ‘infinite variation’ and ‘limited articulation’, enriching the diversity of game world content while maintaining the consistency of the overall scene style [15].

The second is to add intelligent coping strategies for game characters. Characters play a pivotal role in any game. Whether the character object is a character, animal, half-human, god or even mechanical life. As long as they are set as intelligent beings, their character’s decision-making and actions in response to events are especially crucial to the enhancement of game quality. In order to enhance the character of NPC characters, AIGC models can also be used to make NPC characters more flexible and intelligent through machine learning and sentiment analysis, making anthropomorphic and dynamic responses based on the immediate behaviour of the player. This AIGC-driven behavioural design can create more complex and realistic game interaction scenarios, greatly enhancing game immersion. Players will feel that these characters have independent minds rather than just mechanical responses.

Thirdly, to improve the image quality and rendering to enhance the visual impact, AIGC technology through deep learning algorithms for intelligent rendering of game images, without affecting the overall package size of the game, to enhance the image quality of low-resolution images, so that it can achieve or close to the high-resolution naked eye visual effects, indirectly improving the overall image quality of the game [16]. The improvement of the image quality of the game with the same packet size will positively support a higher frame rate of the game, thus reducing the performance demand of the client on the equipment, which is more conducive to the operation and promotion of the game itself. At the same time, the energy consumption required to run the game will also be reduced, making a positive contribution to the global green design.

Fourthly, optimisation and testing. In the process of game development, program optimization and post-testing is also a particularly important part of the AIGC can simulate thousands of types of player behaviour, can automatically detect bugs in the game, saving a lot of manual testing time, through the massive samples can help developers to find traditional methods difficult to find the problem.

Table 2: Input and Output index system

Type	Variable names
Input index	Number of R&D personnel R&D funds
Output index	Revenue from main business Book value of software assets

ai can also optimize the resource management, so that the game in different hardware platforms to run more smoothly. By learning the performance of different devices, AIGC can adjust the resource allocation strategy to ensure that the game can also run smoothly on low-end devices.

4 Measurement of Science and Technology Innovation Efficiency of Chinese Listed Digital Media Companies Based on the DEA Model

4.1 Data sources

The WIND database screened 55 listed companies with AIGC as the subject word. Further, 22 sample companies were selected by using online games, digital media and other keywords as the industry chain screening. To ensure data availability and completeness, 20 sample companies were finally selected. This paper finally analyses the data of these 20 companies for the year 2023. All data analysed are from the WIND financial database.

4.2 Selection and Construction of input-output indicator systems

Selecting the number of R&D personnel and R&D funds as input indicators aligns well with the characteristics of innovation investment, as they directly determine the extent of resources dedicated to innovation, representing the critical resources required for developing and implementing AIGC solutions.

Meanwhile, using revenue from main business and book value of software assets as output indicators effectively reflects the outcomes of technological innovation. Revenue captures the commercial value of AIGC-enhanced products or services, while software assets represent the accumulation of digital intellectual property, such as algorithms, content libraries, or platforms.

See Table 2 for details.

4.3 Introduction of DEA method

The Data Envelopment Analysis (DEA) model is a non-parametric analytical method for efficiency evaluation, which is particularly suitable for multi-input and multi-output scenarios, and is therefore well suited for application in STI efficiency research. Considering the large differences in the size conditions of digital media enterprises, the commonly used BCC model is chosen in the case of Variable Returns to Scale (VRS).

The implication is to assume that there are N companies (which become Decision Making Units, DMU_s), each of which has M input

Table 3: : The result of DEA model comprehensive score

Efficiency Score Range	Efficiency Status	Description	Improvement Suggestions
$\theta = 1$	Fully Effective	Reaches the efficiency frontier with no redundant input or insufficient output	No improvement needed
$0.8 \leq \theta < 1$	Slightly Effective	Close to the efficiency frontier with relatively stable efficiency levels, though some minor resource redundancy or output insufficiency exists	Optimize resource allocation slightly
$0.5 \leq \theta < 0.8$	Moderately Ineffective	Significant gap from the efficiency frontier, with noticeable input redundancy or output insufficiency, leading to lower efficiency levels	Clarify resource use and adjust production output
$\theta < 0.5$	Highly Ineffective	Far from the efficiency frontier with severe resource waste or production insufficiencies	Reevaluate business processes and reallocate resources

indicators and S output indicators. Then for J companies: X_{ij} denotes the value of the i_{th} input indicator for the J_{th} company, and Y_{rj} denotes the value of the r_{th} output indicator for the J_{th} company. The efficiency score of the target company (DMU0) is maximised by determining the optimal input and output weights V_i and U_r .

In the BBC model, variable returns to scale are further introduced by adding the free variable u_0 . This is used to regulate the scale effect, expressed as follows

$$\text{Max}\theta = \sum_{r=1}^s u_r y_{r0} - u_0 \quad (1)$$

The note: θ is the efficiency score of DMU0.

The u_r is the weight of the r_{th} output indicator.

The u_0 is a free variable to control for scale effects.

Meanwhile the constraints are as follows.

$$\sum_{i=1}^m v_i x_{i0} = 1 \quad (2)$$

$$\sum_{r=1}^s u_r y_{rj} - \sum_{i=1}^m v_i x_{ij} - u_0 \leq 0, j = 1, 2, \dots, n \quad (3)$$

$$u_r \geq 0, v_i \geq 0 \quad (4)$$

With these two constraints, it is ensured that efficiency comparisons between different firms are made on the same basis. and that all firms' efficiency scores relative to DMU0 do not exceed 1.

As shown in Table 3, with comprehensive efficiency as the evaluation unit, according to the literature combing, the digital media companies are divided into several efficiency stages.

4.4 Results and analysis

The empirical results show that the average level of technological innovation efficiency of AIGC digital media enterprises is low, only 0.314. But overall the polarisation is more serious. There are 3 enterprises, Jiecheng stock, Xuanya International, Visual China technological efficiency reaches 1. While the gaming enterprise "360" technological efficiency is the lowest in this industry, only 0.018.

Among the 20 companies, 3 of them are CRS, where inputs are proportional to outputs and they are of moderate size. Thirteen

are DRS, where input increases lead to a decrease in the proportion of output increases, and the company needs to downsize or improve efficiency. Four are IRS, Shengtian Networks, Electric Soul Networks, Insight Group, Gomai Culture, where an increase in inputs can lead to a higher proportion of increase in outputs, and the enterprise can further expand its scale.

Each efficiency statistics are specifically shown in Table 4.

The majority of companies (75%) are technically very inefficient, with very few (15%) achieving full efficiency. This indicates that the majority of companies have major problems matching overall resource inputs with outputs. Pure technical efficiency performs slightly better than technical efficiency, with 25% of companies achieving full efficiency, but still 65% perform poorly. This indicates that there are deficiencies in the management level or the use of technology, and companies need to further optimise their operational processes and technical means. In terms of scale efficiency, 40 % of the companies are close to the ideal state, but there are still 25 % of the companies performing moderately ineffective and 20 % highly ineffective. This suggests that some companies may suffer from diseconomies of scale, that is, the size of the company is not suitable for its existing resource allocation.

5 Conclusions and recommendations

75% of companies are highly ineffective in terms of technological efficiency. This is because digital media companies have long technology development cycles and high investment, while short-term outputs are insignificant. This may be due to the high development cost of the AIGC model and the rapid iteration of the technology, which requires significant R&D investment and arithmetic support. The output benefits of content generation are not always evident in the short term.

65 % of companies are highly ineffective in terms of pure technical efficiency. This reflects the general lack of technology application or low management efficiency in companies. This is mainly due to the fact that AIGC needs to be combined with specific business scenarios, but companies may lack efficient scenario-based landing capabilities. The shortage of professional and technical talents or insufficient team synergy affects the effectiveness of generative AI technology application.

Table 4: The result of technological innovation efficiency calculations for 2023

Efficiency Status	technical efficiency		pure technical efficiency		scale efficiency	
	Number of companies	proportion	Number of companies	proportion	Number of companies	proportion
Fully Effective	3	15%	5	25%	3	15%
Slightly Effective	0	0	0	0	8	40%
Moderately Ineffective	2	10%	2	10%	5	25%
Highly Ineffective	15	75%	13	65%	4	20%

55 % of companies are better than lightly efficient in terms of size. This reflects reasonable control of company size, but there are still a small number of companies that are either too large or too small. This is mainly due to the fact that digital media companies need to continuously expand their data resources and user bases, but overexpansion increases the operational burden. Rapid growth brings about an increase in management complexity, which affects the efficiency of intra-enterprise collaboration.

In order to help digital media enterprises make full use of AIGC technology to enhance the efficiency of technology and innovation, the following suggestions are made:

AIGC technology development is costly and requires strong arithmetic support. Companies should optimise resource investment by focusing on high-value scenarios (e.g. personalised recommendation, virtual anchor) and reduce inefficient projects. At the same time, it can combine cloud computing and open source models to reduce arithmetic costs and achieve efficient resource utilisation. In addition, the data-driven feedback mechanism continuously optimises the quality of generated content and user engagement, forming an efficient closed-loop production process.

To improve the efficiency of pure technology, companies should accelerate the application of AIGC in different business scenarios, such as short videos, news content and meta-universe projects. At the same time, they should attract composite talents familiar with AI and digital media to improve the depth of technology usage. With the help of intelligent management platforms, companies can also automate content planning and operation analysis to further enhance synergy efficiency.

For smaller companies, user data and content resources can be expanded through mergers and acquisitions or co-operation; while companies that are too large in scale should split up their business units and form small, independent teams to enhance operational flexibility. In addition, a dynamic monitoring mechanism should be established to adjust the relationship between scale and input-output in a timely manner to ensure that scale growth is synchronised with operational capability.

To enhance market competitiveness, companies should build an open AIGC ecosystem and collaborate with advertisers, content

platforms and game companies to achieve business synergy. At the same time, they should explore diversified realisation models, such as combining virtual IP development, advertising realisation and AIGC-generated content to increase revenue channels [17]. Companies also need to ensure compliance and pay attention to copyright protection and data compliance in the content generation process to avoid potential legal risks.

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